
CLINICAL REPORT

Treatment of Separation Anxiety Disorder with Graduated Exposure and Reinforcement Targeting School Attendance: A Controlled Case Study

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Abstract — A six-year, nine-month old girl with school avoidance and a variety of other anxiety symptoms was diagnosed with Separation Anxiety Disorder (SAD). SAD was treated using graduated exposure with positive reinforcement for approximations of independent school attendance. Treatment focused on systematically shaping increased duration of school attendance while fading mother's proximity and time spent in the classroom and in transit to and from school. A changing criterion single-subject research design was used to demonstrate functional control of the treatment over the child's school attendance. Follow-up indicated that treatment gains were maintained at two- and nine-months post-treatment. SAD is discussed as a childhood disorder involving fear and avoidance of separation from parents in a variety of situations, one of which may be attending school. The importance of differentiating SAD from school phobia and other anxiety disorders is discussed.

For young children, distress and anxiety upon separation from a parent or other attachment figure is developmentally normal and usually transient. When anxiety in response to separation or the anticipation of separation is excessive and persists over time, Separation Anxiety Disorder (SAD) is said to be present. Children diagnosed with SAD avoid separation from their parent (or other attachment figure), frequently express concerns that they or their parent may be harmed when separated, and often panic upon separation (American Psychiatric Association, 1987).

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As is the case with all anxiety-based disorders in children, there is much controversy over SAD as a diagnostic category (Ollendick, 1983; Silverman, 1987; Thyer & Sowers-Hoag, 1988). In particular, SAD has been historically confused with school phobia, making it difficult to interpret much of the research. Recently, a case has been made that SAD and school phobia should be distinguished from one another (Last, Francis et al., 1987; Ollendick & Mayer, 1984; Thyer & Sowers-Hoag, 1988). According to this position, SAD involves fear and avoidance of a variety of situations involving the common theme of separation (with school avoidance being only one of a number of avoided situations). School phobia, on the other hand, involves fear of specific stimuli related to the school situation itself (such as fear of other children at school, ridicule, or criticism by teachers; see Klein & Last, 1989). School avoidance may also be a function of other problems such as depression or conduct disorder, which further supports the position that a careful assessment is critical in formulating the most appropriate intervention.

In addition to the difficulties associated with the failure to differentiate SAD from other problems (such as school phobia or other anxiety disorders), other methodological concerns regarding the treatment literature on SAD have been noted (Klein & Last, 1989). The majority of the research has been conducted using uncontrolled case studies; controlled group and comparative studies are lacking; and the use of objective, standardized assessment procedures to evaluate treatment outcome is infrequent. Thyer and Sowers-Hoag (1988) reviewed the behavioral treatment studies where separation anxiety was "clearly described as a presenting problem" (p. 218). Unfortunately, for many of the cases they describe, it is not entirely clear whether school phobia or other anxieties were present in addition to separation anxiety. According to their review, behavioral treatments for SAD generally employ techniques similar to those used in the treatment of anxiety in adults. Exposure to the feared stimulus is recognized as a critical element in the treatment of anxiety (Marks, 1987). The treatment studies of SAD reviewed by Thyer and Sowers-Hoag (1988) included interventions involving shaping, systematic desensitization, parent training, graduated exposure, and relaxation training. For the majority of behaviorally-based treatments described, exposure to separation was a component of the treatment package. The authors reported positive outcomes for the studies reviewed; however, the majority of studies did not provide adequate experimental control.

In a recent study, Ollendick et al. (1991) reported on the treatment of two cases of SAD using a multiple baseline-across subjects design. For both these cases, which met DSM-III-R criteria for SAD using structured diagnostic interviews, nighttime fears were the primary complaint, whereas school avoidance was not a problem. A reinforcement and self-control training program targeting sleeping apart from their mothers was successful in gradually increasing the childrens' sleeping in their own beds. Although other separation-related anxieties were not targeted by the treatment, neither child met criteria for SAD at post-treatment or follow-up.

Here we report on a young girl for whom school avoidance was the most prominent and disruptive feature of her separation anxiety. Although she had a host of other fears and concerns, school avoidance was selected as the tar-

get of treatment, because this was the most problematic feature of her separation anxiety.

METHOD

Case Description

Jennifer, a six-year, nine-month old white female, had a history of petit mal seizures, which had been well controlled with Zarontin since she was five years old. The presenting concerns were Jennifer's refusal to stay in her classroom without one of her parents present, excessive concerns about her parents' well-being, and concerns about herself becoming ill. Jennifer lived with her parents and two-year old brother, and was enrolled in the first grade. She was verbal, socially skilled, and proved to be an excellent historian during the clinical interviews.

The school avoidance began when she returned to school after an extended holiday break. At that time, Jennifer began reporting numerous somatic concerns prior to leaving for school. Within a few days, she began clinging to her parents, crying, shaking, pleading for them to stay, and chasing after them when they attempted to leave her at school. On one occasion, Jennifer kicked the principal when she attempted to intervene. Eventually, Jennifer's mother (sometimes her father) was taking her to school and staying in her class every day. Jennifer would typically be taken home after a few hours, so her mother could go to work.

According to both Jennifer and her mother, she began expressing concerns about becoming ill approximately six months prior to her first appointment. At that time, Jennifer had a stomach virus, which she presumably caught from her brother. She continued to avoid many of the foods that she had vomited while she was ill. In addition, Jennifer began repeatedly asking her brother if he felt ill every night, sometimes having her mother waking him to ask him how he felt. Jennifer had always slept with her mother and brother (the father worked at night). Although this was not one of the parents' presenting concerns, her mother reported that Jennifer refused to sleep elsewhere.

Upon coming home early from school, Jennifer was allowed to play or watch television. Although she did not have difficulty staying at home with her aunt after leaving school early, she called her mother at work several times a day. She engaged in avoidant behaviors similar to those reported at school when either one of her parents left the house in the evenings or on weekends, and when her mother attempted to take her to Brownies meetings. In discussing her concerns about her mother, Jennifer stated "I always want to be with her in case something happens to her." She reported concerns that her mother might be in an automobile accident, and added that this had occurred two years prior. Her mother acknowledged that she had been in an accident, but indicated that it was very minor. These separation avoidant behaviors had persisted for approximately 10 weeks, until she was first seen in our clinic.

Assessment

A multimodal assessment similar to that described by Hagopian et al. (1990) was conducted. A number of self- and parent-report questionnaires

were administered to both Jennifer and her mother, and behavioral observations were conducted. Separate behavioral and clinical-diagnostic interviews were conducted individually with Jennifer and her mother by a single therapist. During the diagnostic interviews, both mother and child were asked questions about the presence or absence of specific symptoms of the Anxiety Disorders of Childhood or Adolescence, Dysthymia, and Major Depressive Disorder listed in the *Diagnostic and Statistical Manual* (American Psychiatric Association, 1987).

Jennifer met DSM-III-R criteria for Separation Anxiety Disorder (SAD) based on her report and that of her mother, meeting six of the nine DSM-III-R criteria (persistent worry about possible harm befalling mother, refusal to go to school, refusal to go to sleep without mother, physical complaints upon separation, distress in anticipation of separation, distress upon separation; only three are required for a diagnosis). The school avoidance was not a function of a school phobia, since Jennifer did not express fears or concerns about school apart from separating from her parents, and her separation avoidant behavior was not limited to the school situation. As discussed earlier, such differences are viewed as necessary for the differentiation of school phobia from SAD (Last, Francis et al., 1987).

Significant elevations in the Depressed and Somatic Complaints factors were evident on the Child Behavior Checklist (CBCL; Achenbach & Edelbrock, 1983; see Table 1). These elevations were consistent with a number of disorders (e.g., overanxious, major depression, dysthymia — see Achenbach, 1991); however, when considered along with information obtained in the interviews, Jennifer did not meet criteria for these disorders. Ratings on all other factors were in the normal range. Based on the information from the CBCL and interviews, disorders of conduct, attention, and development were ruled out. Jennifer obtained a slightly elevated score of 20 on the Revised Children's Manifest Anxiety Scale (RCMAS; Reynolds & Richmond, 1978). Her score of 6 on the Children's Depression Inventory (CDI; Kovacs, 1978) was low, and was consistent with the interview data, which indicated that depression was not present. Mother's scores on the State-Trait Anxiety Inventory (STAI; Spielberger, 1973) Trait form, and on the Beck Depression Inventory (BDI; Beck et al., 1979) were in the normal range, and family history of anxiety and other psychiatric disorder were not reported.

Pre-Baseline

For the two weeks before the initial session, the mother was instructed to keep a written narrative report of Jennifer's separation avoidant and anxious behaviors when she attempted to leave her at school. She was also instructed to continue to respond to Jennifer during this pre-baseline phase as she had been in the past weeks.

Baseline

During the course of the assessment, a two and one-half week baseline phase was conducted. During this phase, separation was not attempted.

TABLE 1
QUESTIONNAIRE SCORES BY TIME OF ADMINISTRATION

Questionnaires	Time of Administration			
	Treatment		Follow-up	
	Pre	Post	2-month	9-month
Jennifer				
CDI	6	2	0	3
RCMAS (Anxiety)	20	2	0	4
(Lie)	8	8	8	5
Mother				
STAI trait	44	31	29	27
BDI	3	0	0	2
CBCL (raw scores)				
Depressed	14*	3	1	2
Social withdrawal	6	1	1	1
Somatic complaints	9*	2	2	2
Schizoid-obsessive	0	1	0	0
Hyperactive	3	6	4	3
Sex problems	1	0	1	0
Delinquent	1	2	1	0
Aggressive	16	9	7	7
Cruel	1	2	2	0

Note. *denotes a score outside the Normal Range on the CBCL.

Jennifer's mother continued to take Jennifer to school, but remained in class with her for one hour. Mother continued to record Jennifer's avoidant behavior and anxious episodes.

Treatment

Jennifer's refusal to stay in her class without one of her parents present was selected as the primary treatment target because of deterioration in her academic performance and the financial consequences of her mother's absence from work. Furthermore, it was hypothesized that treatment effects in this situation might generalize to the other high-anxiety situations.

Treatment involved graduated exposure to separation from mother with reinforcement for appropriate behavior upon separation. Using a changing criterion design to demonstrate functional control (Barlow & Hersen, 1984), the amount of time Jennifer was required to stay in her class without mother was gradually increased. Criteria for time in class without mother, as well as interaction with and proximity to mother, were faded concurrently. Initially, Jennifer's mother was instructed to take her to school and leave four times during the hour during which they were at school, leaving for only 30 seconds at a time. Over the course of treatment, Jennifer's mother increased the amount of time and distance she was away while Jennifer remained in class.

The bases for the decision to proceed to the next criterion were Jennifer's success in meeting the current criterion at least two times, as well as her own and her mother's level of comfort with proceeding to the next criterion. Mother gradually went from standing outside the door, to walking down the hall, to sitting in her car, to driving to work, to not returning at all, and finally to not taking Jennifer to or from school.

The reinforcement component, which involved providing Jennifer a sticker (after school) for each time she remained in her seat while mother was out of the room (there was no response cost for failed trials), was initiated on the fourth day of treatment. Stickers could be exchanged for prizes at the end of each week. Much praise and encouragement were provided to Jennifer throughout treatment, and positive self-statements were encouraged (although not formally trained). An additional treatment component was implemented at home upon Jennifer's early return from school. She was required to do school work until after lunch, after which time she could play as she wished.

In addition, some general recommendations for dealing with anxious behavior at home were made to Jennifer's parents. They were instructed to provide minimal attention to Jennifer's queries about the safety of foods or about her parents' well-being, and to ignore excessive inappropriate crying. Once Jennifer was attending school more regularly, a reinforcement treatment program was initiated to increase Jennifer's sleeping in her own bed. This procedure involved presenting Jennifer with a sticker and social praise the morning after she successfully slept in her own bed.

RESULTS

For the two weeks prior to their initial session during the pre-baseline phase, mother's written narrative revealed that when she attempted to leave her in her class, Jennifer would "panic" and follow her out into the hallway, often crying, shaking, and pleading for mother to stay. After attempting to leave multiple times per day, Jennifer's mother would eventually take her home. Jennifer immediately followed mother out of class every time she attempted to leave the class, except on two occasions, where Jennifer remained in class for brief periods (she had to be restrained by the teacher on one of these occasions). During the brief periods of time Jennifer was in school without mother, she was reportedly very anxious, and attempted to call her mother.

The amount of time Jennifer spent in class without mother over the course of the assessment and intervention phases is presented in Fig. 1. As instructed, Jennifer's mother did not attempt to leave the class during the baseline phase conducted over the course of the assessment. Jennifer's mother reported that Jennifer did not become anxious, but frequently asked her mother to sit with her and join her in activities.

For the first three days of treatment, Jennifer continued to follow mother immediately each time mother left the classroom. After the addition of the reinforcement component for staying in her seat when mother left, Jennifer remained in class each time mother left. She met the majority of the changing time and proximity criteria over the next six weeks. During the fifth week of treatment, when the criterion was increased to having Jennifer ride the bus to

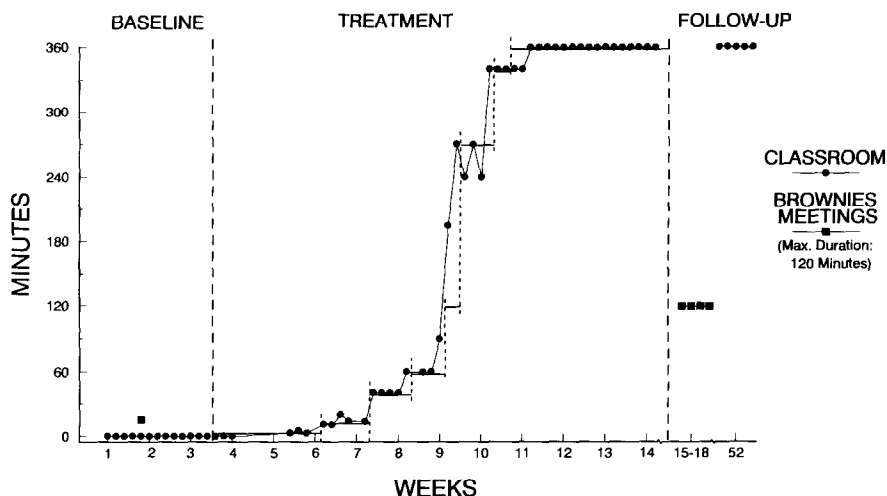


FIG. 1. JENNIFER'S TIME IN CLASS AND BROWNIES MEETINGS WITHOUT MOTHER PRESENT.

school, she refused. After mother rode the bus with her the next two days (without sitting with Jennifer), Jennifer successfully rode the bus without mother thereafter. This was maintained until the end of school year, and the reinforcement component was faded out during the final week.

Following the introduction of the reinforcement program for Jennifer's sleeping in her own bed, improvements were immediately evident. On the first night of treatment, Jennifer was required to stay in her own bed for two hours, but remained the entire night. She did not demonstrate any difficulties sleeping in her own bed thereafter. The stickers were gradually faded and discontinued one week before school ended.

Post-treatment and Follow-up

At post-treatment as well as two- and nine-month follow-up, both Jennifer and mother scored in the normal range on all measures (see Table 1). Jennifer was attending school regularly by the end of treatment, and at her own request, returned to Brownies meetings and attended day camp during the summer. She no longer met any of the DSM-III-R criteria for SAD or any other disorder based on diagnostic interviews conducted with her and her mother separately by the same therapist who conducted the initial assessments and treatment.

At nine-month follow-up, her school attendance during the next academic year remained high, and she continued to participate in Brownies meetings without difficulties (see Fig. 1). During this time, the contingencies present during treatment were not in effect. Jennifer was absent only three days (for legitimate reasons) during each of the 45-day quarters after treatment, compared to the 19 inappropriate absences during the quarter prior to treatment.

Jennifer also continued to sleep in her own bed without difficulty, did not call her mother at work, was eating school and camp lunches (including foods she had previously avoided), and no longer expressed worries about her own or her mother's health. She referred to her past fears and excessive concerns about her mother as the time when "I was acting crazy."

DISCUSSION

The treatment package involving graduated exposure and reinforcement was effective in increasing Jennifer's school attendance and in eliminating her separation anxieties. The changing criterion design demonstrated functional control of the treatment over Jennifer's time in her class without her mother present. Furthermore, the treatment allowed graduated exposure to separation, as well as systematic shaping of both Jennifer's and her mother's behavior.

Given that parents of children with SAD are often anxious themselves (Last, Hersen, Kazdin, Francis, & Grubb, 1987), changing parental behavior may be especially critical in the treatment of SAD in children. For children with SAD, expression of concerns about their parent's (or their own) well-being often results in obtaining reassurance from the parent. Refusal to stay in class or sleep alone, along with crying and pleading, often results in school avoidance or sleeping with the parent, respectively. Indeed, parents of children with SAD often go to great lengths to quell their children's concerns, fears, and sometimes embarrassing avoidant behaviors. These seemingly indulgent parental behaviors may be negatively reinforced by an anxious child becoming calm. Since graduated exposure may be less distressing for both the child and parent, such an approach may be particularly important in the treatment of SAD.

In addition to adding to a limited SAD treatment literature, the present case study demonstrates the utility of a changing criterion design. The changing criterion design may be especially well suited for interventions using graduated exposure to the feared stimulus, since gradual progression is an inherent element of both the design and treatment protocol. In addition, it allows for a systematic demonstration of the effects of reinforcement on shaping approach behavior. Such a design may be more appropriate than reversal designs, since a reversal may interfere with the extinction processes assumed to occur during exposure. Furthermore, in cases where skill acquisition is a component of the treatment package (e.g., parent training, self-instructional strategies), demonstration of functional control using a reversal design may not be possible or appropriate (Barlow & Hersen, 1984).

Although the multimodal assessment using behavioral and clinical diagnostic interviews and standardized questionnaires indicated a diagnosis of SAD, the failure to use a structured diagnostic interview does not allow for the reliable confirmation of the diagnosis of SAD or the exclusion of other diagnoses. Furthermore, it is not possible to rule out experimenter bias, as the same therapist conducted the assessment, treatment, and follow-up. The present study is also limited by the fact that it reports on only a single case, making it difficult to generalize the findings to other individuals, settings, etc. In addition, it is difficult to draw conclusions about the relative importance of the different ele-

ments of the intervention because a formal components analysis was not conducted. Finally, a multiple-baseline across settings design might have been useful in more systematically demonstrating that Jennifer's reduced separation avoidant behavior at school generalized to other settings and situations. Completion of group and comparative studies using standardized assessment instruments and structured diagnostic interviews remains a challenge for researchers examining treatments for SAD and other anxiety-based disorders in children.

Differentiation of SAD from school phobia and other problems that may result in school avoidance is important both in terms of improving the status of the literature and in designing interventions. In the present study, the focus of treatment was on Jennifer's separation from mother. One reason that the school setting was selected was because Jennifer's difficulty separating from her parents at school was the most problematic feature of her separation anxiety. Initially, this may seem to contradict the position that differentiation of SAD from school phobia is important because the treatment described here appears similar to a treatment for school phobia. Had her fears been more related to the school situation itself, however, careful identification of and exposure to those particular stimuli would have been the focus of treatment (see Mansdorf & Lukens, 1987). Similarly, had nighttime problems been more disruptive than school avoidance, sleeping apart from her parents might have been targeted first (see Ollendick et al., 1991). In the present study, and in the two cases described by Ollendick et al. (1991), such an approach appeared to result in the generalized reduction of separation anxieties across other settings and stimuli. Although SAD generally has widespread effects on the child and family, it may be more efficient to focus treatment on the situation in which the separation anxieties are most prominent and disruptive to the child and family.

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